

Board	Component	Part name/Notes	Power Usage - On (W)	End efficiency	Heat Generation - On (W)	Minimum Survival Temperature	Minimum Operating Temperature	Maximum Operating Temperature	Maximum Survival Temperature
Payload	GPR Radar Sounder	REASON-heritage dual-freq sounder	45	0.95	2.25	-40	-30	50	60
EPS	Battery Packs	Li-ion, 2400 Wh (12 h survival)	0	1	0	-40	-20	40	50
EPS	PDU	PCDU, MMRTG EPS heritage	12	0.93	0.84	-30	-20	50	60
Mobility	Drive Motors (4x)	Cryo-rated brushless, MSL heritage	84	0.9	8.4	-40	-30	50	60
GNC	Flight Computer (OBC)	RAD750-class, vault-mounted	18	0.95	0.9	-30	-20	50	60
GNC	I/O & Storage Boards	Logger + ADC stack	6	0.95	0.3	-30	-20	50	60
GNC	Nav Cameras	Stereo VO pair, EECAM heritage	4	0.95	0.2	-40	-30	50	60
Thermal	Heaters	KHLVA-type film heaters	40	1	0				
Thermal	Thermal Sensors	ADT7311-class	0.5	0.95	0.025	-50	-40	85	125
Comms	UHF Radio	Electra-Lite heritage, helix	35	0.95	1.75	-30	-20	50	60
Comms	X-band Transponder	SDST + LGA	45	0.95	2.25	-30	-20	50	60
Total			289.5	0.942	16.91				